



MASTG

0042: Checking for Weaknesses in Third Party Libraries

MASTG-TEST-0043: Memory Corruption Bugs

MASTG-TEST-0044: Make Sure That Free Security Features Are Activated

MASTG-TEST-0222: Position Independent Code (PIC) Not Enabled

MASTG-TEST-0223: Stack Canaries Not Enabled

MASTG-TEST-0245: References to Platform Version APIs

MASTG-TEST-0272: Identify Dependencies with Known Vulnerabilities in the Application

L2

ANDROID

MASWE-0116

TEST



MASTG-TEST-0222: Position Independent Code (PIC) Not Enabled

Overview

This test case checks if the [native libraries](#) of the app are compiled without enabling [Position Independent Code \(PIC\)](#), a common mitigation technique against memory corruption attacks.

Since Android 5.0 (API level 21), Android requires [all dynamically linked executables to support PIE](#).

[Build System Maintainers Guide - Additional Required Arguments](#): Android requires Position-independent executables beginning with API 21. Clang builds PIE executables by default. If invoking the linker directly or not using Clang, use `-pie` when linking.

Steps

1. Extract the app contents ([Exploring the App Package](#)).
2. Run [Obtaining Compiler-Provided Security Features](#) on each shared library and grep for "pic" or the corresponding keyword used by the selected tool.

Observation

The output should list if PIC is enabled or disabled.

Evaluation

The test case fails if PIC is disabled.